221-2327 M. Rohaan Assignment 2

Part - G:-

*Part #01:-

For three nodes x_0, x_1, x_2

$$P_2(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) \dots$$

$c_0 = y[x_0]$, $c_1 = y[x_0 x_1]$, $c_2 = [x_0 x_1 x_2]$.

1) At $x = x_0$

$$P_2(x_0) = c_0 = y_0 \Rightarrow c_0 = y[x_0]$$

2) At $x = x_1$

$$P_2(x_1) = c_0 + c_1(x_1 - x_0) = y_1$$

So,

$$c_1 = \frac{y_1 - y_0}{x_1 - x_0} = y[x_0, x_1]$$

3) At $x = x_2$

$$P_2(x_2) = c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1) = y_2$$

~~$c_2 = (y_2 - y_0 - c_1)(x$~~

$$c_2 = \frac{y_2 - y_0 - c_1(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)}$$

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Substitute $c_1 = \dfrac{y_1 - y_0}{x_1 - x_0}$

$c_0 = y[x_0]$, $c_1 = y[x_0, x_1]$, $c_2 = y[x_0, x_1, x_2]$

$P_2(x) = y[x_0] + y[x_0, x_1](x - x_0) + y[x_0, x_1, x_2](x - x_0)(x - x_1)$

which is the unique quadratic interpolating the 3 points.

*Part #02:-

$$x_i = x_0 + ih, \quad u = \frac{x - x_0}{h}$$

$$P_k(x) = \sum_{m=0}^{k} \frac{\Delta^m y_0}{m!} u(u-1)\cdots(u-m+1)$$

where $\Delta^m y_0$ are forward differences

degree $d \leq k$

* $\Delta^{d+1} p(x) = 0$

$$\Delta^m p(x_0) = 0 \quad \text{for} \quad m > d$$

$$P(x) = \sum_{m=0}^{d} \frac{\Delta^m p(x_0)}{m!} u(u-1)\cdots(u-m+1)$$

and all terms for $m > d$ are zero.

So, $P_k(x) = p(x)$ for all x

Alternatively both $P_k(x)$ and $p(x)$ are Polynomials of degree $\leq k$ and agree at the $k+1$ nodes, so by uniqueness they must be identical.

* The newton forward polynomial of order k reproduces any polynomial of degree $\leq k$ exactly.